UPPER BOUND FOR THE FIBRE DIMENSION AT A POINT WHOSE GENERIC SPECIALISATIONS LIE OUTSIDE THE IMAGE POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof opens with the author's own 'Exactly as in the proof of Lemma ...' and then writes the reduction out; the sentence is reproduced verbatim and the reduction itself is present inside the counted span.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-dimension-fibre-in-higher-codimension is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Dimension formula for a morphism of schemes. Morphisms, Lemma morphisms-lemma-dimension-formula, invoked at proof lines 3686-3688. Supplies the inequality relating the local ring dimension and the transcendence degree.
- Fibre dimension as local ring dimension plus transcendence degree. Morphisms, Lemma morphisms-lemma-dimension-fibre-at-a-point, invoked at proof lines 3692-3694 and 3705. Supplies the identification of d with trdeg_B(A) and the final expression for dim_x(X_y).
- A Noetherian domain mapping onto a quotient with nonzero kernel has strictly larger dimension. Standard algebra, used at proof lines 3695-3698. Supplies the strict inequality between the local ring dimension at x and that at x inside the fibre.

Source reference. Original lemma 0B2I: sources/stacks-varieties/pinned-original-source.tex, source lines 3597–3597. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02FX: sources/stacks-chow/tags.txt, source lines 3208–3208. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 02JU: sources/stacks-chow/tags.txt, source lines 3349–3349. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be locally of finite type. Let $x \in X$ be a point with image $y \in Y$ such that $\mathcal{O}_{Y,y}$ is Noetherian. Let $d \geq 0$ be an integer such that for every generic point η of an irreducible component of X which contains x, we have $f(\eta) \neq y$ and $\dim_\eta(X_{f(\eta)}) = d$. Then $\dim_x(X_y) \leq d + \dim(\mathcal{O}_{Y,y}) - 1$.*

Proof. Exactly as in the proof of Lemma 0B2I we reduce to the case $X = \mathrm{Spec}(A)$ with A a domain and $Y = \mathrm{Spec}(B)$ where B is a Noetherian local ring whose maximal ideal corresponds to y. After replacing B by $B/\operatorname{Ker}(B \to A)$ we may assume that B is a domain and that $B \subset A$. Then we use the dimension formula (Morphisms, Lemma 02JU) to get

$$\dim(\mathcal{O}_{X,x}) + \mathrm{trdeg}_{\kappa(y)}\kappa(x) \leq \dim(B) + \mathrm{trdeg}_B(A)$$

We have $\mathrm{trdeg}_B(A) = d$ by our assumption that $\dim_\eta(X_\xi) = d$, see Morphisms, Lemma 02FX. Since $\mathcal{O}_{X,x} \to \mathcal{O}_{X_y,x}$ has a kernel (as $f(\eta) \neq y$) and since $\mathcal{O}_{X,x}$ is a Noetherian domain we see that $\dim(\mathcal{O}_{X,x}) > \dim(\mathcal{O}_{X_y,x})$. We conclude that

$$\dim_x(X_y) = \dim(\mathcal{O}_{X_y,x}) + \mathrm{trdeg}_{\kappa(y)}\kappa(x) < \dim(B) + d$$

(equality by Morphisms, Lemma 02FX) which proves what we want. □